Blood Cell Classification using CNNs

Kavya Desai | Fall 2025 Curriculum

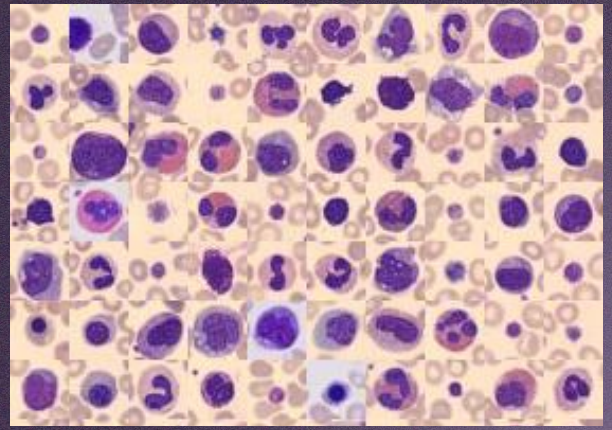
Overview

Goal: Train a CNN to classify microscopic blood smear images into 8 blood cell type categories

Automation ⟶ Faster, more accessible, and more objective

Tools:

- MedMNIST Library for Data
- PyTorch + Torchvision for Deep Learning
- Matplotlib + Seaborn for Visualizations



8 Blood Cell Types	
1. Basophil	5. Lymphocyte
2. Eosinophil	6. Monocyte
3. Erythroblast	7. Neutrophil
4. Immature Granulocyte	8. Platelet

Data

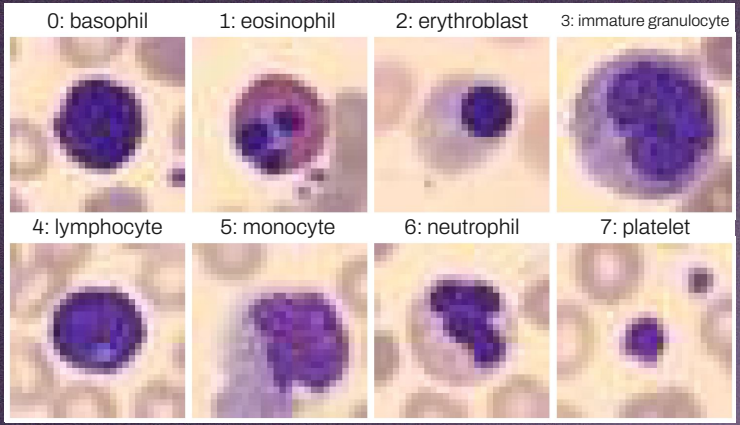
BloodMNIST dataset from the MedMNISTv2 Library

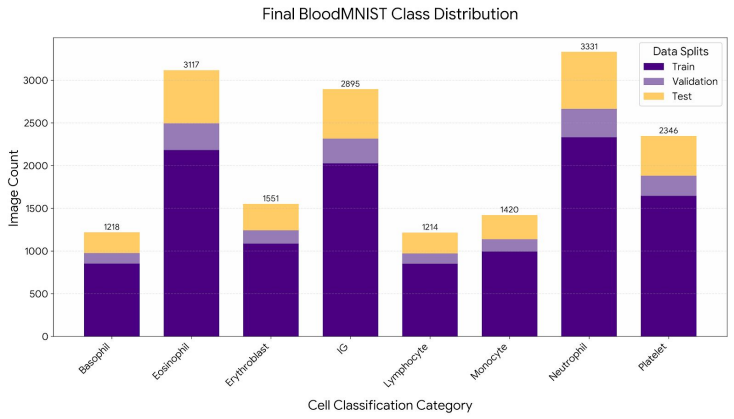
17,092 images
- 11,959 (~70%) Training
- 1,712 (~10%) Validation
- 3,421 (~20%) Testing

28 x 28 resolution, 3-channel (RGB) color space

8 classes, pre-labeled w/ integer label encoding

Pre-processing:
- Stochastic Augmentation (flip/rotate)
- Normalization to [-1, 1]





Model Design

Custom sequential Deep CNN (VGG-inspired)

- 5 convolutional layers (16 filters → 64 filters)
- 3 fully connected layers

Optimizer: Stochastic Gradient Descent (SDG)

- Momentum 0.9
- Learning rate 0.001

Cross Entropy Loss Function

$$L = -\sum_{i=1}^{8} y_i \log(\hat{y}_i)$$

Training

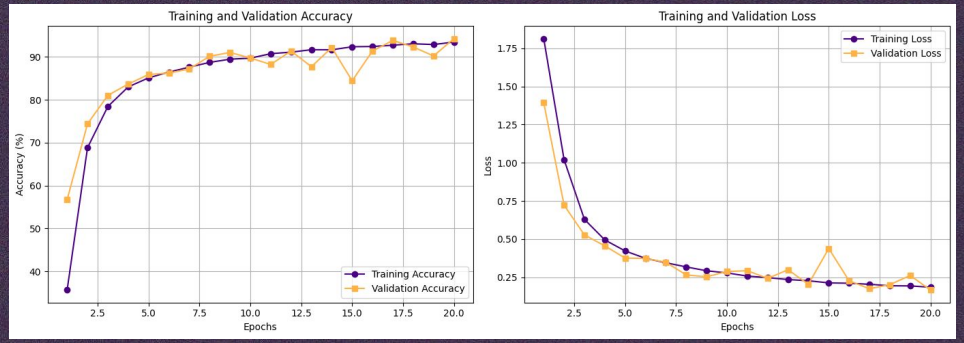
- Batch size: 128
- 20 Epochs

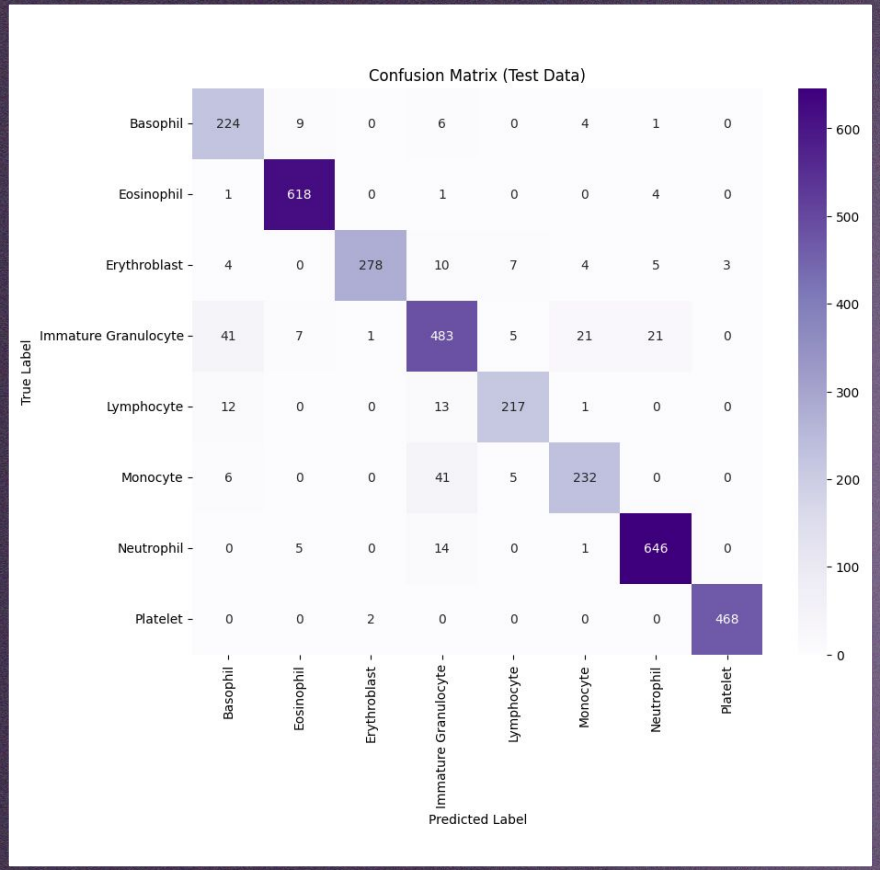
```
self.layer1 = nn.Sequential(
    nn.Conv2d(in_channels, 16, kernel_size=3),
    nn.BatchNorm2d(16),
    nn.ReLU())

self.layer2 = nn.Sequential(
    nn.Conv2d(16, 16, kernel_size=3),
    nn.BatchNorm2d(16),
    nn.ReLU(),
    nn.MaxPool2d(kernel_size=2, stride=2))

self.layer3 = nn.Sequential(
    nn.Conv2d(16, 64, kernel_size=3),
    nn.BatchNorm2d(64),
    nn.ReLU())

self.layer4 = nn.Sequential(
    nn.Conv2d(64, 64, kernel_size=3),
    nn.BatchNorm2d(64),
    nn.ReLU())

self.layer5 = nn.Sequential(
    nn.Conv2d(64, 64, kernel_size=3, padding=1),
    nn.BatchNorm2d(64),
    nn.ReLU(),
    nn.MaxPool2d(kernel_size=2, stride=2))

self.fc = nn.Sequential(
    nn.Linear(64 * 4 * 4, 128),
    nn.ReLU(),
    nn.Linear(128, 128),
    nn.ReLU(),
    nn.Linear(128, num_classes))
```

Model Performance

Final Accuracy: **92.4%** 🥳

```
train  auc: 0.996  acc:0.934
test  auc: 0.995  acc:0.924
```





Next Steps

Train a model with higher-resolution images

Create an interface to deploy the model

Develop end-to-end pipeline for processing microscopic blood smear slides

Input: Wide-Field Microscopic Blood Smear

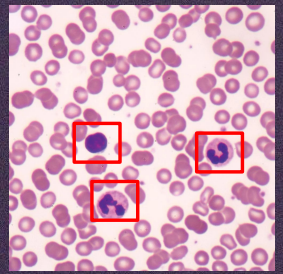
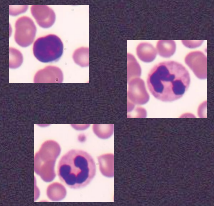


Image segmentation



Pre-processing + Normalization

This project!

CNN Classification

Output: Aggregated Diagnostic Report

White blood cells (total)	• 5000–10,000/μL
Neutrophils	• 55%–70%
Eosinophils	• 1%–3%
Basophils	• 0.5%–1%
Lymphocytes	• 20%–35%
Monocytes	• 3%–8%
Platelets	• 150,000–300,000/μL

Thank You!

Special thanks to Noah, Maddy, and Ben!